

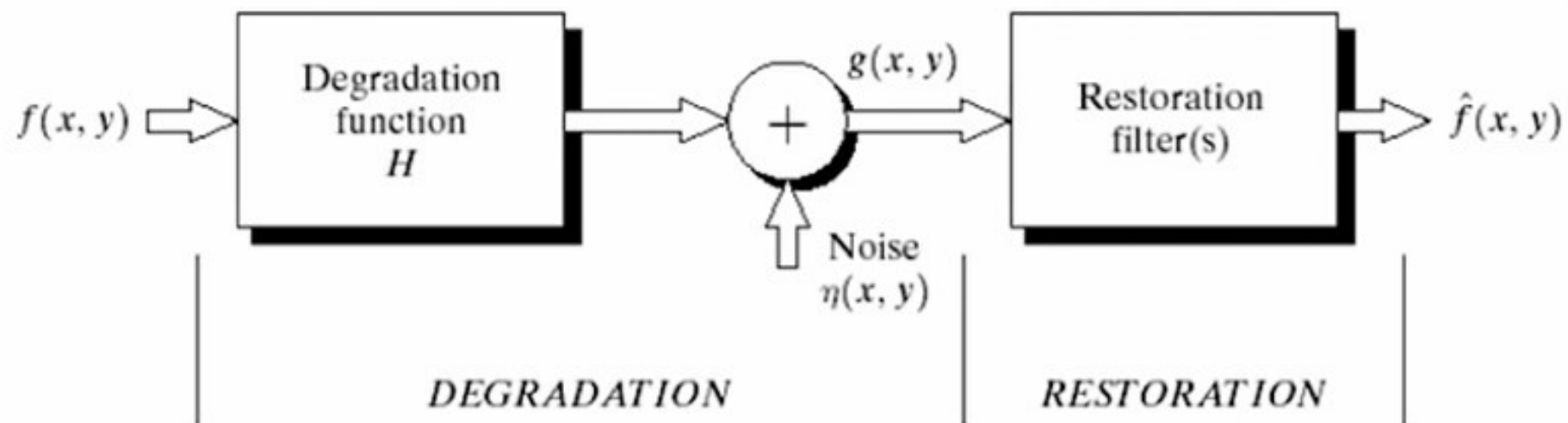
Estimating the degradation function

Topics to be cover

- Estimating the degrading function
 - 1) Estimation by observation
 - 2) Estimation by experimentation
 - 3) Estimation by modelling

What is meaning of image degradation in DIP

- 1) Different ways of Image Degradation
- 2) Various Stages of Image Degradation
- 3) How to estimate the image Degradation



Estimation by Image Observation

- Suppose that we are given a degraded image without any knowledge about the degradation function H .
- One way to estimate this function is to gather information from the image
- For example, if the image is blurred, consider a **small sub image** (**observed sub image**) $g_s(x,y)$, we can manually (filtering) remove the degradation in that region with an estimated sub image.
- Let us assume that the additive noise is negligible in such an area with a strong signal content.
- **constructed sub image** be denoted by $\hat{F}_s(u,v)$

$$H_s(u,v) = \frac{G_s(u,v)}{\hat{F}_s(u,v)}$$

Estimation by Experimentation

- If **equipment** similar to the equipment used to acquire the degraded image is available, it is possible to obtain an accurate estimate of the degradation.
- Images similar to the degraded image can be acquired with various **system settings** until they are degraded as closely as possible to the image. we wish to restore.
- This can be achieved by **applying an impulse (bright dot)** as an input image.
- An impulse is simulated by a bright dot of light, as bright as possible to **reduce the effect of noise**.
- The **Fourier transform of an impulse is a constant**

$$H(u, v) = \frac{G(u, v)}{A}$$

$G(u, v)$: The Fourier transform of the observed image
 A : a constant describing the strength of the impulse

Estimation by Experimentation

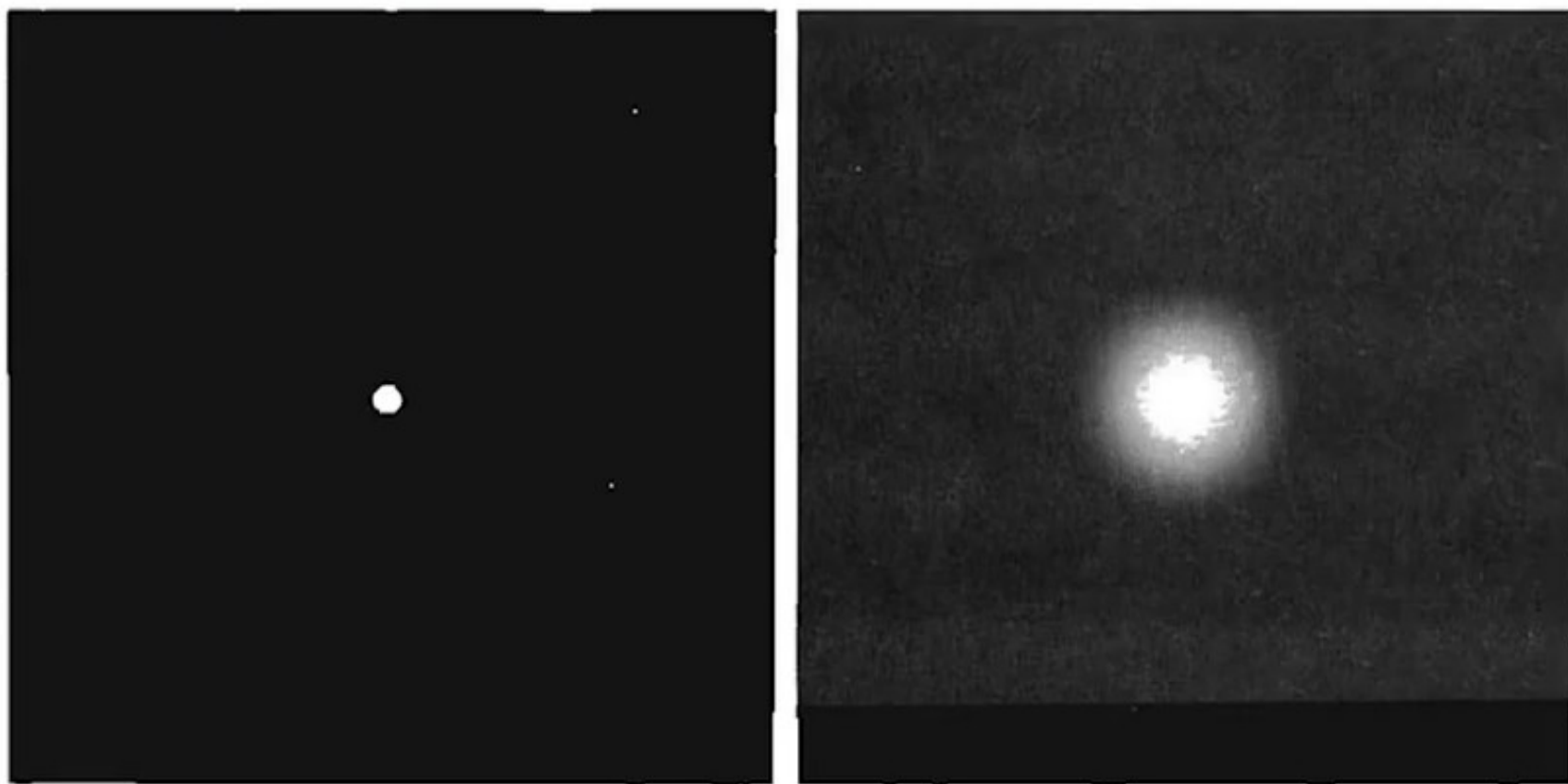


Fig: Degradation estimation by impulse characterization
(a) an impulse of light shown magnified (b) imaged (degraded) impulse

Estimation by Modeling

- A model is a **set of equations** that approximate a real system
- Many types of models have been developed for **modeling degradations** like motion blurs and atmospheric turbulence.
- This model has a Familiar form:

$$H(u, v) = e^{-k(u^2 + v^2)^{5/6}}$$

where

k is a constant that depends on the nature of the turbulence 5/6 expectation value, power on the exponent

k = 0.0025 (in case of severe turbulence)

k = 0.001 (in case of mild turbulence)

k = 0.00025 (in case of low turbulence)

- **Estimation by Modeling**

- Suppose that an image $f(x, y)$ undergoes planar motion and that $x_o(t)$ and $y_o(t)$ are the time varying components of motion in the x- and y-directions,
- T is the duration of the exposure.

- **Then**

$$g(x, y) = \int_0^T f[x - x_o(t), y - y_o(t)] dt \quad \text{--- (2)}$$

where $g(x, y)$ is the blurred image

- From Eq. (1), the Fourier transform of Eq. (2) is

$$\begin{aligned}
 G(u, v) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \underbrace{g(x, y)}_T e^{-j2\pi(ux+vy)} dx dy \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left[\int_0^T f(x-x_0(t), y-y_0(t)) dt \right] e^{-j2\pi(ux+vy)} dx dy \quad \text{--- (3)}
 \end{aligned}$$

- Reversing the order of integration Eq. (3)

$$G(u, v) = \int_0^T \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x-x_0(t), y-y_0(t)) e^{-j2\pi(ux+vy)} dx dy \right] dt \quad \text{--- (4)}$$

- Fourier transform of the displaced function
- Eq. (4) can be written as

$$\begin{aligned}
 G(u, v) &= \int_0^T F(u, v) e^{-j2\pi(ux+vy)} dt \\
 &= \underline{F(u, v)} \int_0^T e^{-j2\pi(ux+vy)} dt \quad \text{--- (5) } \checkmark
 \end{aligned}$$

- $F(u, v)$ is independent of t .

- By defining

$$H(u, v) = \int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]} dt \quad \text{--- (6) }$$

- Eq. (5) may be expressed in the familiar form

$$G(u, v) = H(u, v) \cdot F(u, v) \quad \text{--- (7) }$$

- Comparing Eq. (5) and Eq. (7)

$$H(u, v) = \int_0^T e^{-j2\pi(u x + v y)}$$

Suppose if image undergoes uniform linear motion in x-direction only, at a rate given by $x_o(t) = at/T$. When $t = T$, the image has been displaced by a total distance a . With $y_o(t) = 0$

Eq. (6) gives

$$\begin{aligned} H(u, v) &= \int_0^T e^{-j2\pi u x_o(t)} dt \\ &= \int_0^T e^{-j2\pi u a t / T} dt \\ &= \frac{T}{\pi u a} \sin(\pi u a) e^{-j\pi u a} \quad \text{--- (8)} \end{aligned}$$

- If we allow y component to vary as well then $y_o(t)=bt/T$
- Eq. (8) becomes

$$H(u, v) = \frac{T}{\pi(ua+vb)} \sin(\pi(ua+vb)) e^{-j\pi(ua+vb)} \quad \text{--- (9)}$$

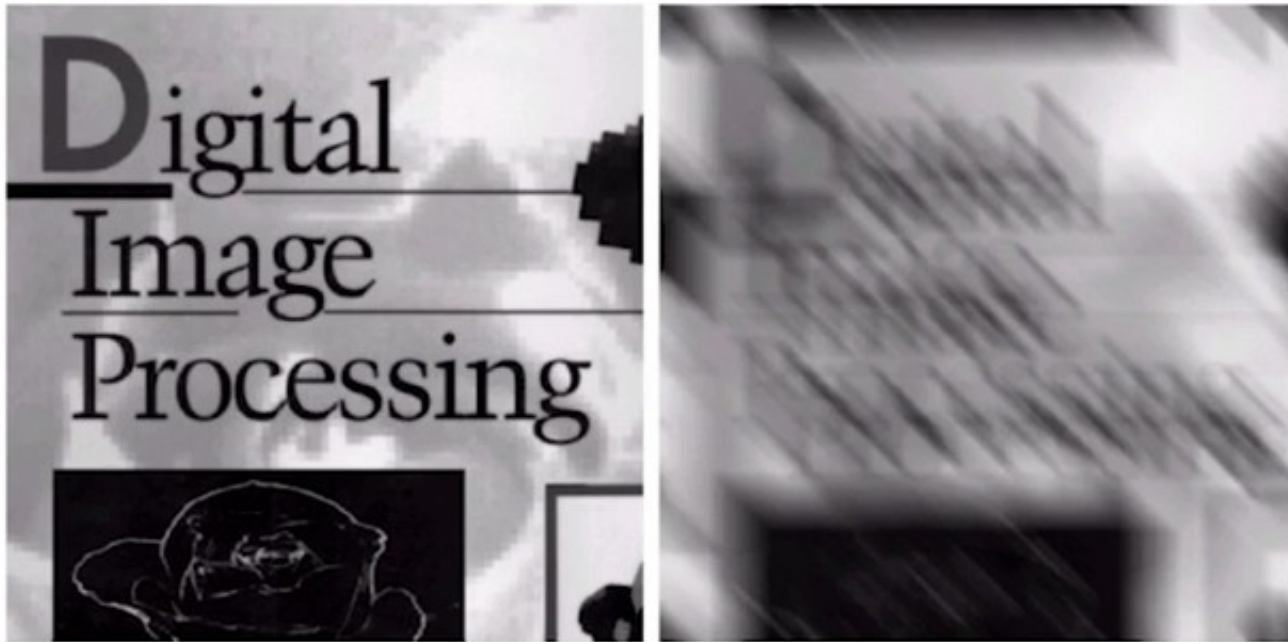


Fig:(a) original image (b) result of blurring with $a=b=0.1$ and $T=1$

Unit -2

Terminology

1. Define Image enhancement.

Image enhancement is the procedure of improving the quality and information content of original image

2. What are primary and secondary colors.

Three Primary Colors: Red, Yellow, Blue.

Three Secondary Colors : Orange, Green, and Violet.

3. What is spatial domain.

The spatial domain is the normal image space, in which a change in position in I directly projects to a change in position in S .

4. What are different noise models in image processing.

Gaussian noise, Rayleigh noise, Erlang (Gamma) noise, Exponential noise, Uniform noise, Impulse (salt-and-pepper) noise.

5. What are the two properties in Linear Operator?

A function “f” is called a linear operator if it has the two properties:

- (i) $f(x+y)=f(x)+f(y)$ for all x and y ;
- (ii) $f(cx)=cf(x)$ for all x and all constants c .

6. How many numbers of steps are involved in image processing?

Image acquisition, Image enhancement, Image restoration, Color image processing, Wavelets and multi resolution processing, Compression, Morphological processing, Segmentation, Representation & description, Object recognition.

7. Write about the concept of algebraic approach?

The concept of algebraic approach is to estimate the original image which minimizes a predefined criterion of performances.

8. Define object recognition.

Object recognition is the process that assigns a label to an object based on its descriptors.

9. What is contrast stretching.

Contrast stretching (often called normalization) is a simple image enhancement technique that attempts to improve the contrast in an image by 'stretching' the range of intensity values it contains to span a desired range of values

11. What is transformation- linear function.

The locations of points (r_1, s_1) and (r_2, s_2) control the shape of the transformation function.

If $r_1 = s_1$ and $r_2 = s_2$ then the transformation is a linear function that produces no changes in gray levels.

12. How transformation becomes a thresholding function.

If $r_1 = r_2$, $s_1 = 0$ and $s_2 = L-1$, the transformation becomes a thresholding function that creates a binary image.

13. Define Gray-level interpolation .

Gray-level interpolation deals with the assignment of gray levels to pixels in the spatially transformed image.

14. Define Smoothing of an image.

Smoothing filters, also known as blurring filters, are a type of image filter that are commonly used in image processing to reduce noise and remove small details from an image.

15. What are different types of Smoothing filters in frequency domain.

Ideal LPF, Butterworth LPF, Gaussian LPF.

16. Define Gray-level slicing .

Highlighting a specific range of gray levels in an image often is desired in gray-level slicing.

17. Define sharpening of an image?

Sharpening filters are a type of image processing that enhances the contrast between neighboring pixels, making the edges and details more visible and defined

18. What is image translation?

Image translation means reposition the image from one co-ordinate location to another along straight line path.

19. What is Image sharpening process?

Image sharpening is an effect applied to digital images to give them a sharper appearance.

20. Define the second order derivative of a digital function.

Must be zero in the flat areas i.e. areas of constant grey values. Must be nonzero at the onset of a gray- level step or ramp discontinuities.

21. List out different types of sharpening filters in frequency domain ?

Ideal HPF, Butterworth HPF, Gaussian HPF

22. What is the purpose of Image differentiation.

Image differentiation enhances the edges, discontinuities and deemphasizes the pixels with slow varying gray levels.

23. What are the three methods of estimating the degradation function?

1. Observation
2. Experimentation
3. Mathematical modeling

24. Define BPF.

A band-pass filter or BPF is a device that passes frequencies within a certain range and rejects frequencies outside that range

25. Define LPF .

A low-pass filter (LPF) attenuates content above a cutoff frequency, allowing lower frequencies to pass through the filter.

26. What is Function of HPF.

A high-pass filter (HPF) attenuates content below a cutoff frequency, allowing higher frequencies to pass through the filter.

27. What is Image negatives?

The negative of an image with gray levels in the range $[0, L-1]$ is obtained by using the negative transformation, which is given by the expression. $s=L-1-r$, Where s is output pixel r is input pixel

29. Give the formula for negative and log transformation.

Negative: $S = L - 1 - r$;

Log: $S = c \log(1+r)$ Where c is constant and greater than or equal to zero

30. Why blur is to be removed from images?

The blur is caused by lens that is improper manner, relative motion between camera and scene and atmospheric turbulence. It will introduce bandwidth reduction and make the image analysis as complex. To prevent the Issues, blur is removed from the images

31. What is meant by Image Restoration.

Restoration attempts to reconstruct or recover an image that has been degraded by using a clear knowledge of the degrading phenomenon.

32. List the two properties in Linear Operator?

Additivity and Homogeneity

33. Define homogeneity property in Linear Operator.

The homogeneity property says that ,the response to a constant multiple of any input is equal to the response to that Input multiplied by the same constant.
